Abstract

- An app designed for color blindness to coordinate and choose clothes with confidence and elegance. Recognizing the challenges that color blind patients face in the world of fashion, our app uses the latest technology and user-centered design to provide a seamless and comprehensive solution**.**
- Help with coordinating clothes: By providing clothing coordination suggestions based on color compatibility and fashion trends.
- Users can choose to add a photo from their computer to our app. In the process of adding pictures, this picture will be automatically recognized by our pre-trained model at one time, by type (such as T-shirt), gender, color, season, usage (such as formal, casual, etc.)

Overview

CBRO is a mobile application designed to assist colorblind individuals in identifying colors accurately.

With approximately 300 million people worldwide affected by colorblind. CBRO aims to provide a practical solution to everyday challenges faced by colorblind individuals, particularly in tasks such as selecting clothing, matching colors, and identifying element color.

Utilizing image recognition and data processing techniques, the application achieves high precision and effectiveness in color recognition. Our goal is to enhance the lives of users affected by color blindness, providing a comprehensive and user-friendly solution to improve their daily experiences.

Problem Definition

Colorblind individuals face difficulties in accurately identifying and distinguishing colors, which can lead to challenges in tasks such as selecting clothing, matching colors, and identifying item color. This can make everyday tasks harder and cause feelings of inconvenience, leading to challenges in daily life.

Color blindness is usually inherited. Men are more likely to be born with color blindness. Most people with color blindness can't distinguish between certain shades of red and green. Less commonly, people with color blindness can't distinguish between shades of blue and yellow.

Understanding Color Blindness

Color blindness, also known as color vision deficiency, is a visual impairment that affects an individual's ability to perceive certain colors accurately. This condition can range from mild to severe and can be either inherited or acquired later in life due to various factors.

Types and Impacts on Vision

There are several types of color blindness, the most common of which include:

Red-Green Color Blindness: This is the most prevalent type of color vision deficiency. Individuals with red-green color blindness have difficulty differentiating between shades of red and green.

Blue-Yellow Color Blindness: This type of color blindness, also known as tritanopia, affects the perception of blue and yellow colors. Individuals with tritanopia have difficulty distinguishing between shades of blue and yellow.

Complete Color Blindness: This rare form of color blindness, known as monochromacy or achromatopsia, results in the inability to perceive any colors. Individuals with complete color blindness see the world in shades of gray.

Normal Vision Monochromacy

Project Goal

The primary objective of this project is to develop an inclusive mobile application that caters to the unique challenges faced by colorblind individuals in selecting and coordinating clothing. By leveraging advanced algorithms and intuitive design, the app aims to provide accurate color identification, contrast analysis, and personalized outfit recommendations. Ultimately, the goal is to empower colorblind users to confidently express their personal style and navigate the world of fashion with ease.

App Features

- **Virtual Closet:** Stores user's clothes for easy outfit planning.
- Outfit Creator: Generates outfit ideas based on the user's clothes and preferences.
- **Easy-to-Use Interface:** Designed to be accessible for colorblind users.
- **Educational Features:** Provides tips on color coordination and using the app effectively.

- **Personalized Suggestions:** Recommends clothes based on user preferences and any color vision issues.
- **Color Identification:** A tool that accurately recognizes colors in clothing items.

Purpose

These documents aim to explain our purpose in developing this application, which targets a specific group of users. The main objective is to provide an enhanced user experience for individuals with color blindness, enabling color recognition and offering tools for easier clothing coordination. Notable features include color recognition, clothing coordination, and assistance for the color-blind, allowing them a more integrated experience with their surroundings.

We hope that this application will significantly improve the lives of individuals facing challenges in color perception, and we continuously strive to deliver a unique and beneficial experience.